
Online Examination Report

Date:	21.07.2014
Attempts:	1
Examinee:	Shyaka, Daniel
Date of birth:	5/26/1993
Subject:	080-Principles of Flight
Result:	32,49 %
Time used:	00:00:13 (HH:MM:SS)

Attachments

1. List of result by question
2. List of result by topic
3. Detailed question result report

Attachment 1: List of result by question

CQB Number	Internal Number	Syllabus reference	Chapter	Points	Max. Points
1	185	81. PRINCIPLES OF FLIGHT AEROPLANE		1,00	1,00
2	189	81. PRINCIPLES OF FLIGHT AEROPLANE		0,00	1,00
3	194	81. PRINCIPLES OF FLIGHT AEROPLANE		0,00	1,00
4	195	81. PRINCIPLES OF FLIGHT AEROPLANE		1,00	1,00
5	197	81. PRINCIPLES OF FLIGHT AEROPLANE		1,00	1,00
6	211	81. PRINCIPLES OF FLIGHT AEROPLANE		0,00	1,00
7	239	81. PRINCIPLES OF FLIGHT AEROPLANE		0,00	1,00
8	180	81. PRINCIPLES OF FLIGHT AEROPLANE		0,00	1,00
9	217	81. PRINCIPLES OF FLIGHT AEROPLANE		0,00	1,00
10	210	81. PRINCIPLES OF FLIGHT AEROPLANE		0,00	1,00
11	147	81. PRINCIPLES OF FLIGHT AEROPLANE		0,00	1,00
12	214	81. PRINCIPLES OF FLIGHT AEROPLANE		0,00	1,00
13	37	81.1.1.3 Aerodynamic forces and moments on aerofoils		1,00	1,00
14	17	81.1.1.5 Wing shape		0,00	1,00
15	23	81.1.2.4 Centre of pressure and aerodynamic centre		0,00	1,00
16	100	81.1.3.1 The lift coefficient C_l		1,00	1,00
17	84	81.1.3.1 The lift coefficient C_l		1,00	1,00
18	74	81.1.4.2 Induced drag		1,00	1,00
19	34	81.1.4.2 Induced drag		0,00	1,00
20	70	81.1.4.2 Induced drag		0,00	1,00
21	90	81.1.4.2 Induced drag		1,00	1,00
22	54	81.1.4.2 Induced drag		0,00	1,00
23	35	81.1.6 Ground effect		0,00	1,00
24	12	81.1.8.2 The stall speed		1,00	1,00
25	38	81.1.8.2 The stall speed		1,00	1,00
26	78	81.1.12.1 Ice and other contaminants		0,00	1,00
27	52	81.2.2.1 Normal shock waves		1,00	1,00
28	39	81.2.3.4 Effect on pitching moment		0,00	1,00

CQB Number	Internal Number	Syllabus reference	Chapter	Points	Max. Points
29	102	81.4	STABILITY	1,00	1,00
30	101	81.4	STABILITY	0,00	1,00
31	14	81.4.1.1	Basics and definitions	0,00	1,00
32	86	81.4.1.1	Basics and definitions	0,00	1,00
33	15	81.4.4.4	Factors affecting static directional stability	0,00	1,00
34	98	81.5.8.3	Stabiliser trim	0,00	1,00
35	96	81.6.1.1	Flutter	0,00	1,00
36	109	81.7	PROPELLERS	0,00	1,00
37	107	81.7	PROPELLERS	0,00	1,00
38	64	81.7.4.1	Torque reaction	1,00	1,00
39	3	81.8.1.5	Steady coordinated turn	0,00	1,00
40	97	81.8.1.5	Steady coordinated turn	0,00	1,00
Total:32%				13,00	40,00

Attachment 2: List of result by topic

Licence

Date: 18.07.2014

Examinee: Shyaka, Daniel

Location: Uganda Civil Aviation Authority

Your Result: 13,00 from 40,00 Points (32,50 %)

Topics	Amount of Questions	Points achieved	Maximum Points	Percent
81. PRINCIPLES OF FLIGHT AEROPLANE	40	13,00	40,00	32,50
81.1. SUBSONIC AERODYNAMICS	14	7,00	14,00	50,00
81.2. HIGH SPEED AERODYNAMICS	2	1,00	2,00	50,00
81.4. STABILITY	5	1,00	5,00	20,00
81.5. CONTROL	1	0,00	1,00	0,00
81.6. LIMITATIONS	1	0,00	1,00	0,00
81.7. PROPELLERS	3	1,00	3,00	33,33
81.8. FLIGHT MECHANICS	2	0,00	2,00	0,00
Total	40	13,00	40,00	32,50

Candidate: Daniel Shyaka
Date of birth: 1993-05-26

Subject: 080-Principles of Flight
Date: 2014-07-18



1

If the landing gear on an airplane moves forward during retraction, the total moment will

☐ increase.

☐ remain the same.

☒ decrease.

2

When one engine fails on a twin-engine airplane, the resulting performance loss

- ☒ is approximately 50 percent since 50 percent of the normally available thrust is lost.
- ☐ reduces cruise indicated airspeed by 50 percent or more.
- ☐ may reduce the rate of climb by 80 percent or more.

3

If the CG of an aircraft is moved from the aft limit to beyond the forward limit, how will it affect the cruising and stalling speed?

- ☒ **Increase both the cruising speed and stalling speed.**
- ☐ **Decrease the cruising speed and increase the stalling speed.**
- ☐ **Decrease both the cruising speed and stalling speed.**

4

When an aircraft's forward CG limit is exceeded, it will affect the flight characteristics of the aircraft by producing

- ☐ very light elevator control forces which make it easy to inadvertently overstress the aircraft.
- ☒ higher stalling speeds and more longitudinal stability.
- ☐ improved performance since it reduces the induced drag.

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If the gyroplane's CG is below the propeller thrust line, which direction will the application of power cause the nose to move?

☒ **The nose will pitch down.**

☐ **The nose will pitch up.**

☐ **The nose will not move.**

6

If swirling dust, leaves, or debris indicate a strong thermal on the final approach to a landing, it is recommended that the glider pilot

- ☐ open the spoilers and reduce the airspeed.
- ☒ open the spoilers and maintain a constant airspeed.
- ☐ close the spoilers and increase the airspeed.

7

(Refer to Figure 145.) What is the correct sequence for recovery from the unusual attitude indicated?

See Attachments:

1. instrument_145.jpg

figure 245

(Attachment
No.1)

- ☐ Reduce power, level the wings, bring pitch attitude to level flight.
- ☒ Level the wings, raise the nose of the aircraft to level flight attitude, and obtain desired airspeed.
- ☐ Reduce power, increase back elevator pressure, and level the wings.

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8

Which type of flap is characterized by large increases in lift coefficient with minimum changes in drag?

☐ **Fowler.**

☐ **Split.**

☒ **Slotted.**

9

What is the relationship of false lift to the wind? False lift

- ☒ increases if the vertical velocity of a balloon increases.
- ☐ exists only if the surface winds are calm.
- ☐ decreases as the wind accelerates a balloon to the same speed as the wind.

10

A rule of thumb for flying a final approach is to maintain a speed that is

- ☐ 50 percent above the glider's stall speed plus half the estimated windspeed.
- ☒ twice the glider's stall speed plus half the estimated windspeed.
- ☐ 50 percent above the glider's stall speed, regardless of windspeed.

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When a slight upward or negative flap deflection is used, the result is

☒ decreased drag.

☐ decreased lift.

☒ increased drag.

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12

What is the suggested speed to fly when passing through lift with no intention to work the lift?

- ☒ **Best glide speed.**
- ☐ **Best lift/drag speed.**
- ☐ **Minimum sink speed.**

13

The angle between the direction of the undisturbed airflow (relative wind) and the chord line of an aerofoil is called:

- ☒ angle of attack.
- ☐ climb path angle.
- ☐ same as the angle between chord line and fuselage axis.
- ☐ glide path angle.

14

Taper ratio of a wing is the ratio between:

- ☒ tip chord and root chord.
- ☐ mean geometric chord and wing span.
- ☐ wing span squared and wing area.
- ☒ root chord and tip chord.

15

Assuming no flow separation and no compressibility effects the location of the centre of pressure of a symmetrical aerofoil section:

- ☒ is at approximately 25% chord irrespective of angle of attack.
- ☐ is at approximately 50% chord irrespective of angle of attack.
- ☐ moves backward when the angle of attack decreases.
- ☒ moves forward when the angle of attack decreases.

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16

(Please view annex 081-0006 issue date July 2004)

The point in the annex corresponding to CL for minimum horizontal flight speed is:

☐ point a.

☐ point d.

☒ point c.

☐ point b.

See Attachments:

1. 081-0006.gif

Annex

(Attachment
No.2)

17

The lift formula can be written as:
(rho = density)

☐ $L = n * W.$

☒ $L = CL * \frac{1}{2} \rho * V^2 * S.$

☐ $L = CL * 2\rho * V^2 * S.$

☐ $L = W.$

18

An aeroplane transitions from steady straight and level flight into a horizontal co-ordinated turn with a load factor of 2, the speed remains constant and the:

- ☐ total drag increases by a factor of 4.
- ☒ induced drag increases by a factor of 4.
- ☐ lift increases by a factor of 4.
- ☐ angle of attack increases by a factor of 1/4.

19

Which of these statements about induced drag are correct or incorrect?

- I. An elliptical spanwise lift distribution generates less induced drag than a rectangular lift distribution.
- II. Induced drag decreases with decreasing aspect ratio.

☐ I is incorrect, II is incorrect.

☐ I is correct, II is incorrect.

☒ I is incorrect, II is correct.

☐ I is correct, II is correct.

20

Which of the following wing shapes produces the lowest induced drag? (assume zero wing twist)

☐ Rectangular.

☐ Circular.

☐ Elliptical.

☒ Tapered.

21

Excluding constants, the coefficient of induced drag (CDi) is the ratio of:

- ☐ CL and CD.
- ☐ CL^2 and S (wing surface).
- ☒ CL^2 and AR (aspect ratio).
- ☐ CL and b (wing span).

22

Winglets:

- ☒ create an elliptical lift distribution.
- ☐ decrease the induced drag.
- ☐ increase the manoeuvrability.
- ☐ decrease the static lateral stability.

23

Which statement, about an aeroplane leaving ground effect at constant angle of attack, is correct?

- I. The lift coefficient C_L remains constant.
II. The induced drag coefficient C_{Di} decreases.

- ☒ I is incorrect, II is correct.
- ☐ I is correct, II is correct.
- ☐ I is incorrect, II is incorrect.
- ☐ I is correct, II is incorrect.

24

The stall speed:

- ☒ increases with an increased weight.
- ☐ does not depend on weight.
- ☐ increases with the length of the wingspan.
- ☐ decreases with an increased weight.

25

The stall speed increases, when: (all other factors of importance being constant)

- ☐ weight decreases.
- ☐ spoilers are retracted.
- ☐ minor altitude changes occur e.g. 0-10.000 ft.
- ☒ pulling out of a dive.

26

The effects of very heavy rain (tropical rain) on the aerodynamic characteristics of an aeroplane are:

- ☐ decrease of CLMAX and decrease of drag.
- ☒ increase of CLMAX and decrease of drag.
- ☐ increase of CLMAX and increase of drag.
- ☐ decrease of CLMAX and increase of drag.

27

**Which of the following statements is correct?
Air passes a normal shock wave as**

☒ **The static temperature increases.**

☐ **The static temperature decreases.**

☐ **The velocity increases.**

☐ **The static pressure decreases.**

28

Which of these statements about "tuck under" are correct or incorrect?

- I. A contributing factor to "tuck under" is a forward movement of the centre of pressure of the wing.
- II. A contributing factor to "tuck under" is an increase in the downwash angle at the location of the horizontal stabiliser.

- ☐ I is correct, II is correct.
- ☐ I is incorrect, II is incorrect.
- ☒ I is incorrect, II is correct.
- ☐ I is correct, II is incorrect.

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29

The capability of an aircraft to respond to a pilot's inputs, especially with regard to flightpath and attitude, is

☒ controllability.

☐ maneuverability.

☐ response.

30

If the aircraft's nose initially tends to move farther from its original position after the elevator control is pressed forward and released, the aircraft displays

☐ positive static stability.

☐ negative static stability.

☒ positive dynamic stability.

31

A statically unstable aeroplane:

- ☒ reduces its instability as the CG moves aft.
- ☐ can never be dynamically stable.
- ☐ will become stable at high speed.
- ☐ is always dynamically stable.

32

Positive static stability of an aeroplane means that following a disturbance from the equilibrium condition:

- ☐ the tendency is to move with an oscillatory motion of decreasing amplitude.
- ☒ the initial tendency is to diverge further from its equilibrium condition.
- ☐ the initial tendency is to return towards its equilibrium condition.
- ☐ the tendency is to move with an oscillatory motion of increasing amplitude.

33

Which of the following provides a positive contribution to static directional stability?

- ☐ A forward swept wing.
- ☐ A low wing as compared with a high wing.
- ☐ A dorsal fin.
- ☒ Moving the centre of gravity aft.

34

What is the effect on landing speed when a trimmable horizontal stabiliser jams at high IAS?

- ☒ No effect when landing on a high elevation runway.
- ☐ In most cases, no effect.
- ☐ No effect with a forward CG.
- ☐ In most cases, a higher than normal landing speed is required.

35

Wing flutter may be caused by a:

- ☐ combination roll control reversal and low speed stall.
- ☐ combination of fuselage bending and wing torsion.
- ☒ combination of bending and torsion of the wing structure.
- ☒ aerodynamic wing stall at high speed.

36

A propeller rotating clockwise, as seen from the rear, creates a spiraling slipstream that tends to rotate the aircraft to the

- ☒ left around the vertical axis, and to the left around the longitudinal axis.
- ☐ left around the vertical axis, and to the right around the longitudinal axis.
- ☐ right around the vertical axis, and to the left around the longitudinal axis.

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37

The distance a propeller actually advances in one revolution is

☐ effective pitch.

☐ geometric pitch.

☒ twisting.

38

The torque reaction of a rotating fixed pitch propeller will be greatest at:

- ☐ low aeroplane speed and low engine power.
- ☒ low aeroplane speed and maximum engine power.
- ☐ high aeroplane speed and low engine power.
- ☐ high aeroplane speed and maximum engine power.

39

An aeroplane performs a steady horizontal, co-ordinated turn with 45 degrees of bank at 230 kt TAS. The same aeroplane with the same bank angle and speed, but at a lower mass will turn with:

- ☐ a larger turn radius.
- ☐ a lower rate of turn.
- ☒ a smaller turn radius.
- ☐ the same rate of turn.

40

An aeroplane performs a steady horizontal, co-ordinated turn with 45 degrees of bank at 230 kt TAS. The same aeroplane with the same bank angle and speed, but at a lower mass will turn with:

- ☐ the same turn radius.
- ☒ a smaller turn radius.
- ☐ a higher rate of turn.
- ☐ a larger turn radius.

Attachment No. 1

Questions 7

figure 245

instrument_145.jpg

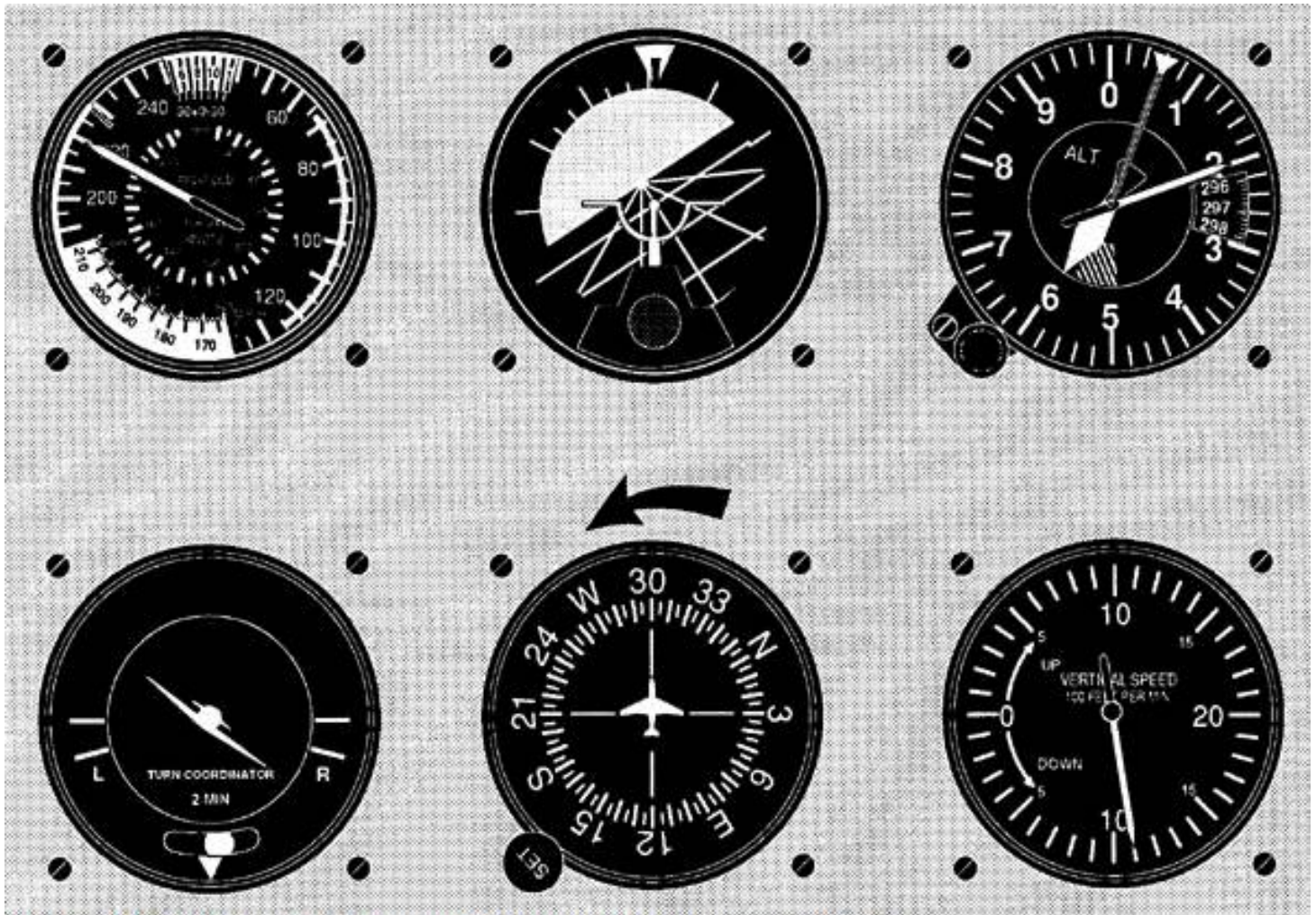


Figure 145. Instrument Sequence (Unusual Attitude) © ASA

Attachment No. 2

Questions 16

Annex

081-0006.gif

081 - 0006

